Non-generic continuation: an ordered-root constructor obligation

2026-10-26. The remaining task is a uniform constructor for an order-sensitive coordinate predicate. This note defines that object, gives its reduction to scalar recovery, and supplies a cheap wrong-root control. It also resolves the parameter-dependence caveat in a Miller differential screen. These are derived mathematical checks and bounded source inspection; no new scientific run or algorithmic speedup was measured.

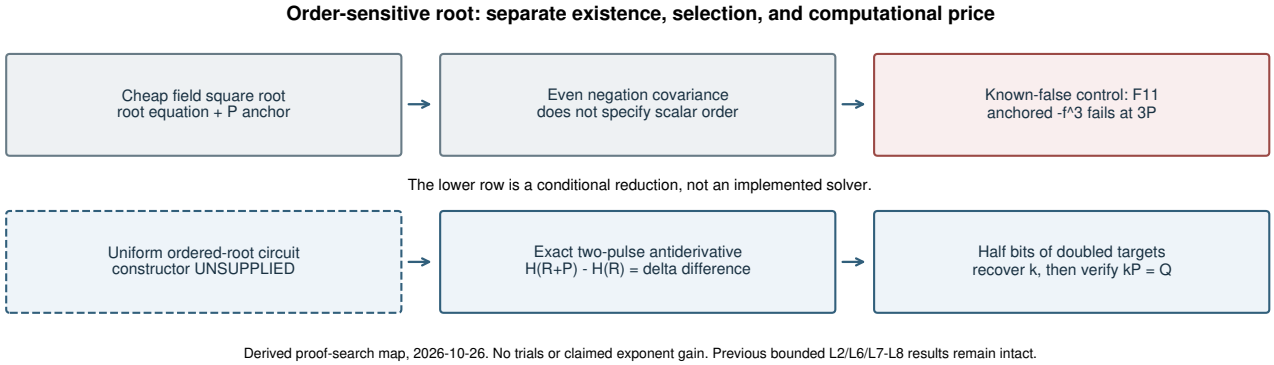

Figure 1: Analytical proof-search map. The missing constructor is explicit; existence, a correct field root, and a short symbolic expression are not a priced evaluator.

Exact object and the sparse difference equation

Let $E : y^2 = f(x) = x^3 + ax + b$ over $\mathbf{F}_p$, with $p > 3$, and let P have odd prime order $r \neq p$. Put $m = (r-1)/2$. The subgroup abscissas $x(jP)$, $1 \leq j \leq m$, are distinct. Define

$$K_P(X) = \prod_{j=1}^{m} (X - x(jP)), \qquad A_P(x(jP)) = y(jP), \quad \deg A_P < m.$$

Interpolation gives a unique A_P, and $A_P^2 = f \pmod{K_P}$. More importantly,

$$y(kP) = A_P(x(kP)) \quad \Longleftrightarrow \quad 1 \leq k \leq m.$$

Thus this *generator-ordered* square root would decide a scalar half, rather than an arbitrary point-sign convention.

Equivalently, let $H(kP) = 0$ for $0 \leq k \leq m$ and $H(kP) = 1$ for $m+1 \leq k < r$. It is the unique solution on the subgroup of

$$H(O) = 0, \qquad H(R+P) - H(R) = \delta_{mP}(R) - \delta_{-P}(R).$$

The two pulse locations are public. A finite-point equality indicator uses Fermat zero tests with $O(\log p)$ field multiplications; identity points are handled separately. The missing operation is a cheap, uniform **discrete antiderivative** of this sparse right-hand side in the finite coordinate algebra. Iterating around the orbit gives it in linear work and is not the desired construction.

The values $H(Q), H(2Q), \ldots$ give the binary digits of k/r. For $L = \lceil \log_2 r \rceil + 1$ digits representing the integer v, the interval

$$v/2^L \leq k/r < (v+1)/2^L$$

contains one possible scalar, $k = \lceil rv/2^L \rceil$, which is publicly checked. This is a reduction conditional on the constructor, not an implemented solver.

Four cheap audits

Identifiability. The congruence $A^2 = f \pmod{K_P}$ has exactly 2^m roots: each abscissa permits an independent sign. The kernel polynomial is unchanged when P is replaced by any other generator. Only the specified ordering selects the desired root. Correct roots, subgroup membership, or one anchored value do not identify it.

Quantifier and price. The required statement is one uniform procedure from public (E, P, r), producing a circuit correct for all targets in the declared family. Construction, constants, kernel data, exceptional inputs, arithmetic bit cost, evaluation and final recovery all count. Existence by interpolation, a circuit built from known logs, or target-specific advice is not this procedure.

Nearby known-false control. For $p = 3 \bmod 4$, $A_0 = f^{(p+1)/4}$ is a short circuit and a valid square root on the subgroup abscissas. Normalize it at P:

$$\sigma_P(X) = \frac{y(P)}{A_0(x(P))} A_0(X).$$

It even has $\sigma_P(x(P)) = y(P)$ and $\sigma_{-P} = -\sigma_P$, but need not have scalar ordering. On $E/\mathbf{F}_{11} : y^2 = x^3 + x + 6$, there are 13 points. With $P = (2,7)$, $2P = (5,2)$ and $3P = (8,3)$, the normalized circuit is $-f^3$. It gives $\sigma_P(8) = 8 \neq 3$, although $3 \leq m = 6$. This hand-derived counterexample is a required control, not an experiment.

Method ceiling. For $r \geq 11$, the nonzero polynomial $A_P^2 - f$ has at least m roots, hence $\deg A_P \geq \lceil m/2 \rceil$. Dense coefficients therefore require linear-size output. This does not lower-bound straight-line circuit size: A_0 already demonstrates high degree with a short circuit. Likewise, a short functional DAG for a translated sum must not treat calls at different points as one evaluation. An expanded sum of s translated two-pulse functions has support at most $2s$, so this elementary representation requires linear terms; nonlinear coordinate circuits remain outside that statement.

Relationship to previous work

The pilot's L2 measured linear EDS complexity and one 15-monomial quadratic grammar; L6 enumerated residual states and orientation data; L7/L8 used truth-derived coefficients. None supplied a uniform high-degree circuit constructor for this exact ordered root. Their bounded results do not establish a circuit-size lower bound.

The atlas's low-degree coordinate census, torsor/deck rational-section obstruction, and recognizable/invertible-subset lead are relevant constraints. Here the root identity holds only modulo a finite subgroup polynomial, not as a rational section of the whole elliptic double cover. The half predicate is not a homomorphism or a translation-respecting small quotient. The precise degree-versus-circuit obligation is therefore not discharged by those arguments. Nevertheless, earlier scalar-orbit period and root-to-source orientation proposals already expose related constructor gaps: different wording is not evidence of novelty.

Current scope: no qualified new constructor was found in this search. This is a sharpened proof obligation, not an admitted experimental algorithm. The next mathematical submission must provide its explicit circuit and cost recurrence before another cohort is collected for it.

Miller differential: preserve the variable-parameter constant

Fix $S \neq O$ in the rational r-torsion subgroup and normalize the Miller function with divisor $r[S] - r[O]$. For $t = -x/y$ write

$$f_{r,S} = t^{-r}(1 + C_r(S)t + \cdots), \qquad \omega = dx/(2y).$$

Riemann-Roch gives the identity in the **evaluation point**

$$\frac{df_{r,S}/f_{r,S}}{\omega} = r\frac{y + y_S}{x - x_S} + C_r(S).$$

For fixed base S, the evaluation-point dependence is a chord slope. But if the Miller parameter is the variable target $S = Q$, the nonlinear public quantity $C_r(Q)$ remains and must not be discarded.

Normalize line/vertical factors $g_{R,S}$ to leading coefficient one. Their first Laurent coefficient is the chord/tangent slope $\lambda(R,S)$; for vertical or identity addition take $\lambda = 0$. Divisors and leading terms give

$$C_r(R+S) = C_r(R) + C_r(S) - r\lambda(R,S).$$

Set $C_r(O) = 0$. Therefore $Z = -C_r/r$ is the unique primitive satisfying

$$Z(R+S) = Z(R) + Z(S) + \lambda(R,S).$$

Uniqueness follows because the difference of two primitives is a homomorphism from an r-group to the additive p-group. A normalized Miller-chain coefficient recurrence computes $C_r(S)$ in $O(\log r)$ group/field steps. These identities were checked in a separate read-only algebra pass; this was not an independent scientific review.

This identifies the map with **IDEA-20261016-09fd08**, the existing zeta-defect proposal. Its proposed chance/white-spectrum predictions are not established by that proposal's existence. The earlier IDEA-20260905-24b41a also groups Miller and additive torsor coordinates together. No new scalar-information claim or family closure follows here.

A model boundary also remains explicit: KN-TECH-005 applies to opaque group encodings. KN-FIND-002 labels its coordinate-based jet discussion a structured GGM and acknowledges that the opaque simulator lacks coordinate access. A lower-bound transfer between those models needs its own argument; it is not supplied by calling a coordinate function deterministic.

Provenance and next discriminating step

Read sources: the completed pilot report; the exotic EDS, coordinate-statistic, p-adic, torsor, Miller and invertible-subset atlas rows; the exact Miller/zeta proposals; KN-FIND-002 and KN-TECH-005. The current source hashes and revisions are in source-manifest.json. Targeted searches across the three repositories and public literature did not locate a matching ordered-root constructor. This bounded search is not a novelty or impossibility result; the knowledge-index tools were unavailable.

Require the next proposed constructor to pass the anchored $\mathbf{F}_{11}$ control, the exact two-pulse identity and its point exceptions, generator/coordinate covariance, and a charged construction/evaluation recurrence. A failed candidate only resolves its stated grammar. A proved small circuit advances implementation readiness; it does not by itself establish a runtime exponent or move a scientific ledger status. The linked problem draft retains this missing ingredient and its exit conditions.